

The Double Cap Conjecture

OPG-60000

L1 - TRACTABLE

PARTIALLY SOLVED

Combinatorics

ORIGINAL PROBLEM STATEMENT

Conjecture The largest measure of a Lebesgue measurable subset of the unit sphere of $\mathbb{R}^n$ containing no pair of orthogonal vectors is attained by two open caps of geodesic radius $\pi/4$ around the north and south poles.

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>